

Stochastic Simulation - Assignment 1

Bitcoin Mempool

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1 Introduction

In this assignment, we simulate the Bitcoin mempool: the pool of unconfirmed transactions waiting to be included in a block by a miner. Unlike a standard FIFO queue, the miner selects transactions greedily by fee-per-byte (highest first) subject to a block size limit. At time 0 the mempool is empty.

2 Event types

The simulation contains three types of events: transaction arrivals, block mining events, replace-by-fee events.

2.1 Transaction arrival

For a transaction n , we compute its size, fee, fee rate according to deterministic formulas. We insert transaction to the mempool and schedule a future RBF event 180 seconds later, which only activates if the transaction is still not confirmed yet.

2.2 Block mining

A block mining event occurs every 600 seconds. We sort the transactions in according to the descending order of its fees, we use transaction ID as the tie breaker. We select the transactions until it exceeds the block capacity of 1,000,000 bytes. Once the transactions that are selected are confirmed, we remove it from the mempool.

2.3 Replace-by-fee

Every transaction has an RBF event scheduled 180 seconds after arrival. If the transaction is not confirmed by that time or if it has not already been bumped, then we increase the fee by 50%. Only bumping the fee one time if it's ever done. We ignore this event if it is already confirmed.

3 Flowchart

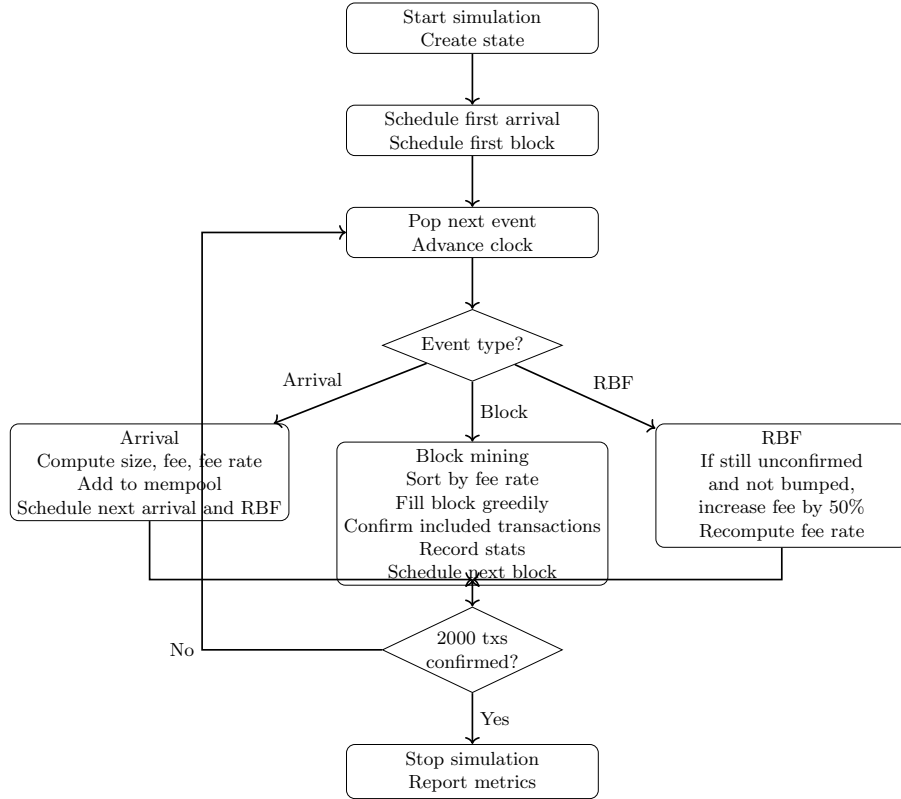


Figure 1: Flowchart of the Bitcoin mempool discrete-event simulation.

4 Mempool priority ordering and RBF

The mempool is represented as a list of pending transactions. So at every mining event, we sort the list into descending order of the fee rate. RBF increases the fee by 50%. The transaction is updated in-place. Since we order the list in descending order every time, the bumped transactions automatically get the new priority.

5 Interaction between confirmation and RBF events

If a transaction is confirmed before its RBF event, then the RBF event of the transaction is ignored. However, if an RBF event occurs but the transaction is

not confirmed, nor bumped previously, then we get the 50% bump. So RBF event is always executed, but it has no effect if it is no longer valid.

6 Results

Metric	Value
Average confirmation time, all transactions	302.51 s
Average confirmation time, RBF transactions	393.22 s
Average confirmation time, non-RBF transactions	93.22 s
Average mempool size	18.90 transactions
Average block utilisation	0.0099

We can see that the average mempool size was around 18.90 transactions. So only a small number of transactions were waiting at any given time. We can also see that the block utilisation was only 0.99%. So the block capacity is much larger than the demand created by the transactions. RBF transactions have a higher average confirmation time as expected, as they needed to have waited for a lot longer in order to get the bump.